

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, JULY 2023
Third / Fourth Semester

21CSC206T - ARTIFICIAL INTELLIGENCE

(For the candidates admitted from the academic year 2021-2022 & 2022-2023)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Marks BL CO PO

Answer ALL Questions

- | | | | | |
|---|---|---|---|---|
| 1. _____ is the common language for AI | 1 | 1 | 1 | 1 |
| (A) Lisp | | | | |
| (B) Python | | | | |
| (C) PHP | | | | |
| (D) Java | | | | |
| 2. The PEAS in the task environment is about _____ | 1 | 1 | 1 | 1 |
| (A) Peer, Environment, Actuators, Sense | | | | |
| (B) Perceiving, Environment, Actuators, Sensors | | | | |
| (C) Performance, Environment, Actuators, Sensors | | | | |
| (D) Perceiving, Environment, Actuators, Sense | | | | |
| 3. In state-space, the set of actions for a given problem is expressed by the _____ | 1 | 1 | 1 | 1 |
| (A) Intermediate States | | | | |
| (B) Successor function that takes current action and returns next state | | | | |
| (C) Initial States | | | | |
| (D) Action sequence | | | | |
| 4. A self-driving car is an example of _____ environment | 1 | 1 | 1 | 1 |
| (A) Static | | | | |
| (B) Known | | | | |
| (C) Discrete | | | | |
| (D) Continuous | | | | |
| 5. The number of edges from the root to the node is called _____ of the tree. | 1 | 1 | 2 | 1 |
| (A) Height | | | | |
| (B) Depth | | | | |
| (C) Length | | | | |
| (D) Width | | | | |
| 6. The time and space complexity of BFS is (For time and space complexity problems consider b as branching factor and d as depth of the search tree.) | 1 | 1 | 2 | 1 |
| (A) $O(bd+1)$ and $O(bd+1)$ | | | | |
| (B) $O(b^2)$ and $O(d^2)$ | | | | |
| (C) $O(d^2)$ and $O(b^2)$ | | | | |
| (D) $O(d^2)$ and $O(d^2)$ | | | | |
| 7. The heuristic function $h(n)$ is which of the following? | 1 | 1 | 2 | 1 |
| (A) Lowest path cost | | | | |
| (B) The cheapest path from the root to the goal node | | | | |
| (C) Average path cost | | | | |
| (D) The estimated cost of the cheapest path from the root node to the goal node | | | | |

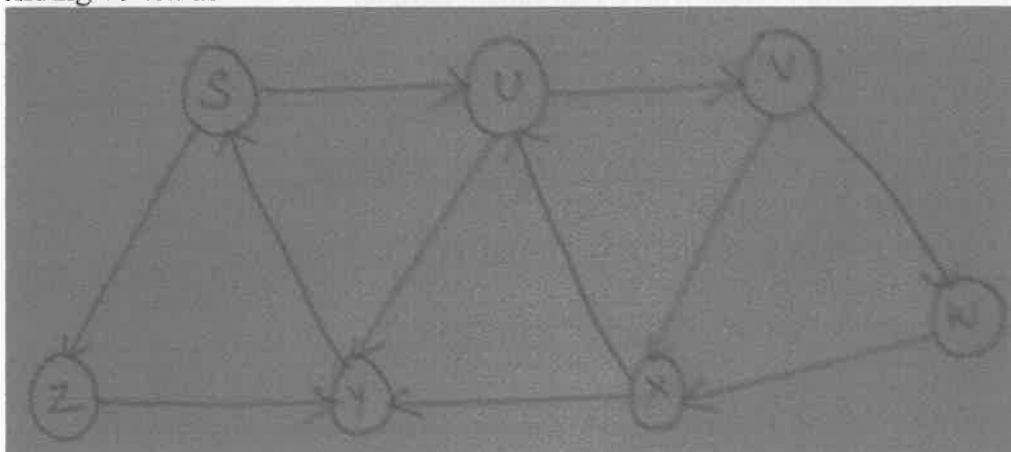
8. Which of the following is the evaluation function in A* method? 1 1 2 1
 (A) Path cost from starting node to the present node (B) Path cost from starting node to the present node and Heuristic cost
 (C) Heuristic function (D) Average of Path cost from starting node to the present node and Heuristic cost
9. For alpha-beta pruning is which option is correct 1 1 3 1
 (A) $\alpha \leq \beta$ (B) $\alpha \geq \beta$
 (C) $\alpha = \beta$ (D) $\alpha \neq \beta$
10. Select the appropriate technique that can be used for Map coloring problem 1 1 3 1
 (A) Breadth first search (B) A* search
 (C) Means-end analysis (D) Constraint satisfaction
11. Agents behavior can be best described by _____. 1 1 3 1
 (A) Perception sequence (B) Sensors and Actuators
 (C) Agent function (D) Environment in which agent is performing
12. Treatment chosen by doctor for a patient for a disease is based on _____. 1 1 3 1
 (A) Only current symptoms (B) Current symptoms plus some knowledge from the textbooks
 (C) knowledge from the experience (D) Current symptoms plus some knowledge from the textbooks plus experience
13. Inference algorithm is complete only if _____. 1 1 4 1
 (A) It can derive any sentence (B) It can derive any sentence that is an entailed version
 (C) It is truth preserving (D) It can derive any sentence that is an entailed version & It is truth preserving
14. The CORRECT formula for the sentence, "not all rainy days are cold" is 1 2 4 1
 (A) $\forall d (\text{Rainy}(d) \wedge \neg \text{Cold}(d))$
 (B) $\forall d (\neg \text{Rainy}(d) \rightarrow \text{Cold}(d))$
 (C) $\exists d (\neg \text{Rainy}(d) \rightarrow \text{Cold}(d))$
 (D) $\exists d (\text{Rainy}(d) \wedge \neg \text{Cold}(d))$
 (A) A (B) B
 (C) C (D) D
15. Choose the correct option. 1 1 4 1
 A) Knowledge base (KB) is consists of set of statements.
 B) Inference is deriving a n
 (A) A is true, B is true (B) A is false, B is false
 (C) A is true, B is false (D) A is false, B is true

16. Wumpus World is a classic problem, best example of _____.
 (A) Single player Game (B) Two player Game
 (C) Reasoning with Knowledge (D) Knowledge based Game
17. Suppose we want to eliminate the inaccuracy problem in partial-order planning problem or planning problem, then the best data structure to use is the
 (A) Stacks (B) Planning Graphs
 (C) BST (Binary Search Tree) (D) Queue
18. Which of the following are the two major characteristics which combine the AI Planning problem?
 (A) FOL and Logic (B) Logic and Knowledge Based Systems
 (C) Knowledge Based Systems (D) Search and Logic
19. The first expert system was _____.
 (A) DENDRAL (B) MYCIN
 (C) XCON (D) CON
20. Machine learning is a subset of which of the following?
 (A) Deep Learning (B) Data Learning
 (C) Artificial Intelligence (D) Big data

PART – B (5 × 8 = 40 Marks)
 Answer ALL Questions

Marks BL CO PO

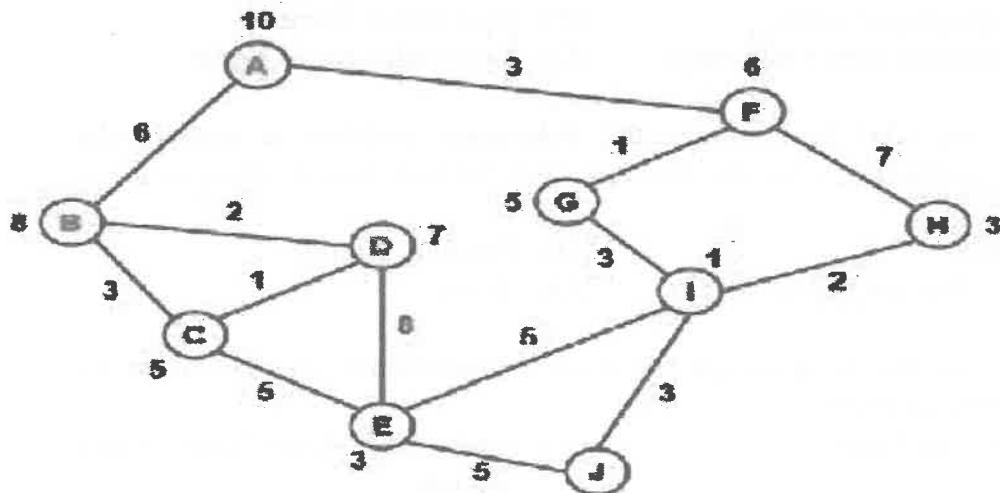
21. a. Explain the various AI models with an example.
- (OR)
- b. Examine the complete solution for Missionaries and Cannibals problem.
22. a. Identify the data structure used in Breadth First Search algorithm (BFS). Apply BFS for the given directed graph with starting vertex as "S" and ending vertex as "X".



(OR)

b. Consider the following graph

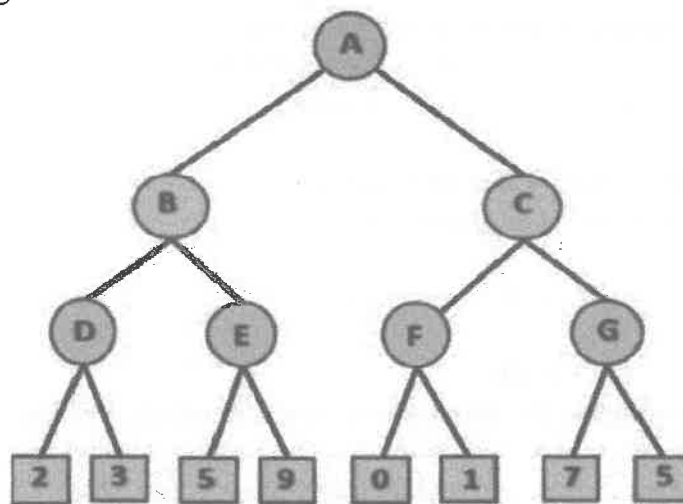
8 3 2 2



The numbers written on edges represent the distance between the nodes. The numbers written on nodes represent the heuristic value. Find the most cost-effective path to reach from start state A to final state J using A* Algorithm.

23. a. Demonstrate the two-player search tree to understand the working of Alpha-beta pruning.

8 2 3 2



(OR)

b. Examine the multi-agent system with a real time example.

8 2 3 2

24. a. Consider the following axioms. Represent these axioms in predicate calculus.

8 3 4 2

- (i) Every child love Ice-cream
- (ii) Everyone who loves Ice-cream loves any chocolate
- (iii) Apple is a fruit, and Apple is red color
- (iv) Anything which has a red nose is weird or is a clown
- (v) No dog is a clown
- (vi) John does not love anything which is weird.

(OR)

b. The law says that "it is a crime for an American to sell weapons to hostile nations. The country "Nano", an enemy of America has some missiles, and all of its missiles were sold by colonel west, who is an American". Use resolution principle to prove that west is a criminal.

8 3 4 2

25. a. Explain the Mean Ends analysis with an example.

8 2 5 1

(OR)

b. Describe the architecture of expert systems.

8 2 5 1

PART – C ($1 \times 15 = 15$ Marks)

Marks BL CO PO

Answer ANY ONE Questions

26. Given two water jugs with capacities X and Y litres. Initially, both the jugs are empty. Also given that there is an infinite amount of water available.

15 4 1 3

The jugs do not have markings to measure smaller quantities.

One can perform the following operations on the jug:

- Fill any of the jugs completely with water.
- Pour water from one jug to the other until one of the jugs is either empty or full, $(X, Y) \rightarrow (X - d, Y + d)$
- Empty any of the jugs

The task is to determine whether it is possible to measure Z litres of water using both the jugs. And if true, print any of the possible ways.

27. Consider the following facts:

15 3 4 3

Prove the conclusion using Resolution.

a. Team Delhi

b. Team Mumbai

c. Final match between Delhi and Mumbai

d. Delhi scored 350 runs, Mumbai scored 350 runs, Delhi lost 5 wickets, Mumbai lost 7 wickets.

e. The team which scored the maximum runs wins

f. If the scores are same the team which lost minimum wickets wins the match.

Represent the facts in predicate, convert to clause form and prove by resolution "Delhi wins the match".

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